Wavefront-Based Visual Acuity Estimation Using Machine Learning

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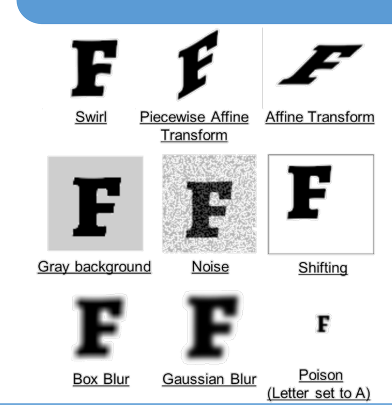
Introduction

ARA, Inc. and AFRL developed RECOIL (Ronchi Evaluator and Classifier of Imperfect Lenses), a Python-based machine learning application, to ensure prototype protective eyewear developed by the United States Air Force (USAF) meets MIL-DTL-43511D for acceptable optical distortion. RECOIL 1.3 uses an AI-powered virtual acuity exam to predict whether lenses will maintain the critical 20/20 visual acuity required for aircrew, significantly accelerating the evaluation process. This guarantees that only lenses meeting both the distortion and USAF vision standards are met.

Methods

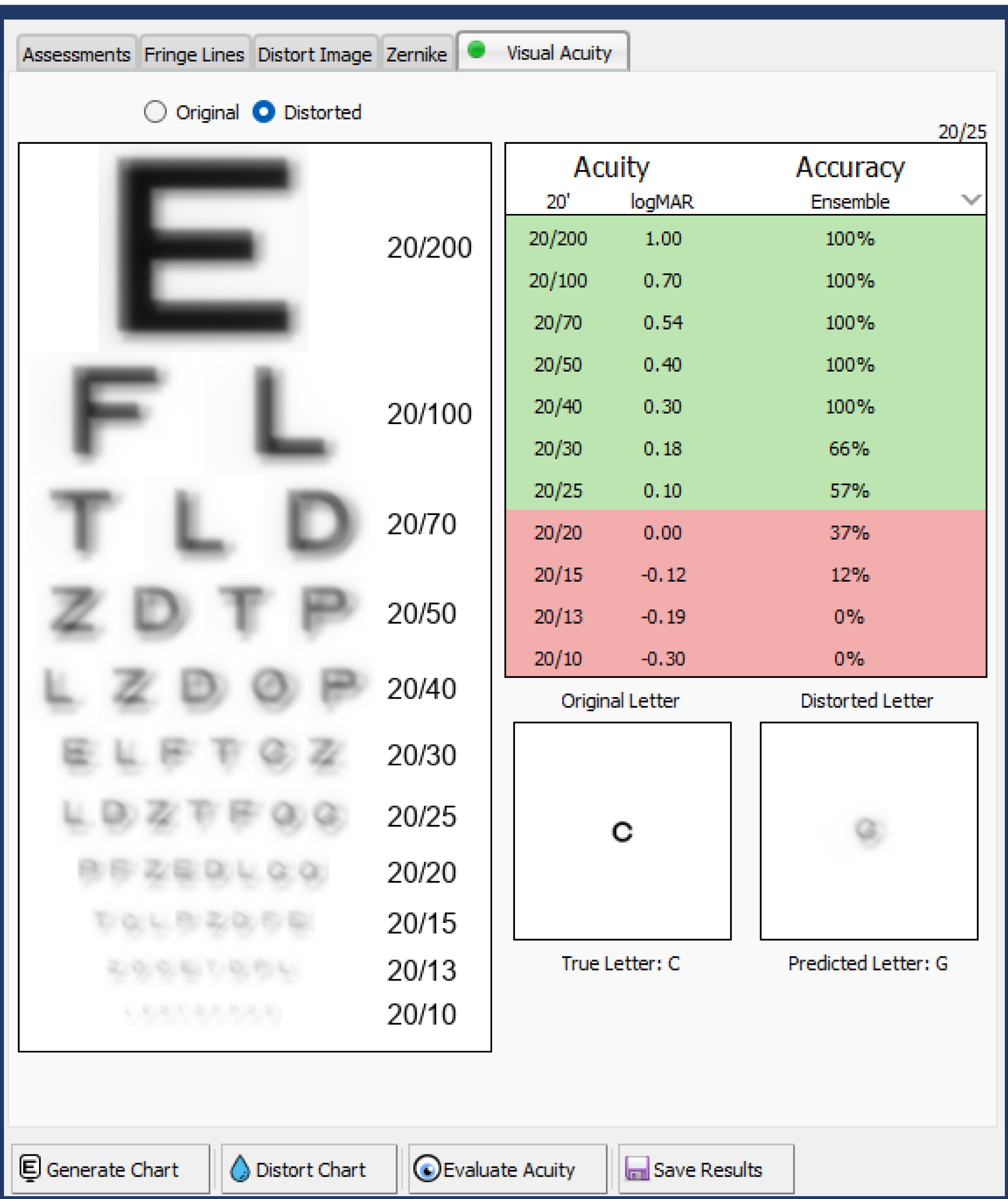
- Computer vision algorithms (using OpenCV, Shapely, & SciPy Python packages) isolate fringes within interferograms, e.g., Ronchigrams
- Fringe Tracing & fast Fourier transforms fit aberration coefficients to Zernike polynomials to mathematically define the wavemap and interferogram pattern
- The Prysm Python library simulates an image scene as viewed through a lens leveraging Zernike polynomials and their coefficients
- GIMP is used to generate font images for training classifier models

Machine Learning



- Several deep, convolutional neural networks (CNN) were developed using TensorFlow 2.17 & PyTorch 2.7.1 to provide a variety of machine learning classifier models to represent subjects with 20/20 vision
- One model combined a CNN with XGBoost (enhanced classification), and another was based on YOLO V7
- Models were trained using dozens of highly disparate fonts – all 26 capital letters with and without anti-aliasing and augmentation (i.e., distortions) and sizes across all lines in a Snellen Chart (**left**)
- Models must accurately classify most letters to progress through the eye chart like a human (**right**)
- Data poisoning and intentional unbalancing techniques ensure model performance plummets at 20/20 acuity
- The virtual Snellen test was used to evaluate visual acuity for thousands of Zernike polynomial coefficients (**ZPCs**) – XGBoost was used to create a direct ZPC-to-visual acuity estimation model

Acuity Line	# of Correct Letters
20/200	26
20/100	≥25
20/70	≥25
20/50	≥24
20/40	≥24
20/30	≥23
20/25	≥23
20/20	≥22
20/15	≤3
20/13	≤3
20/10	≤3

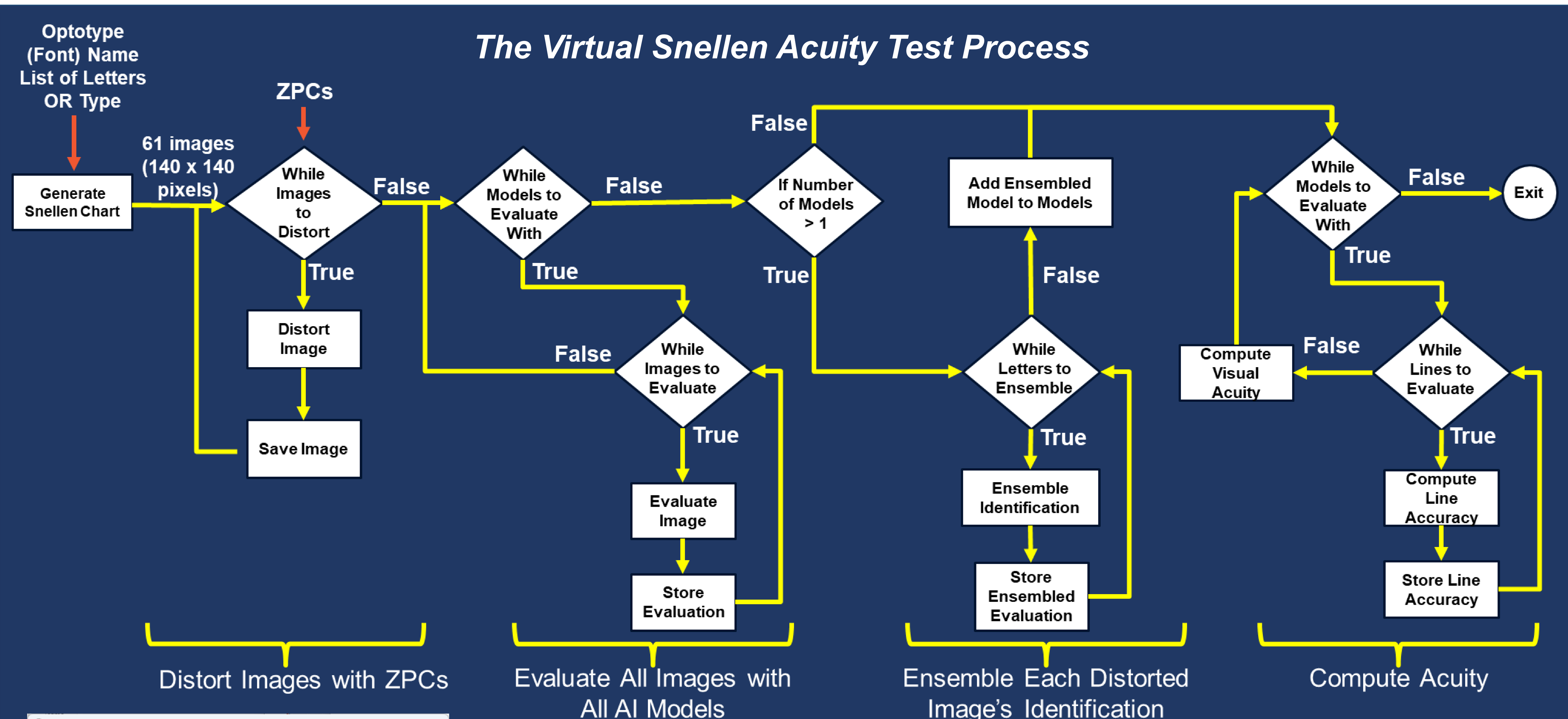


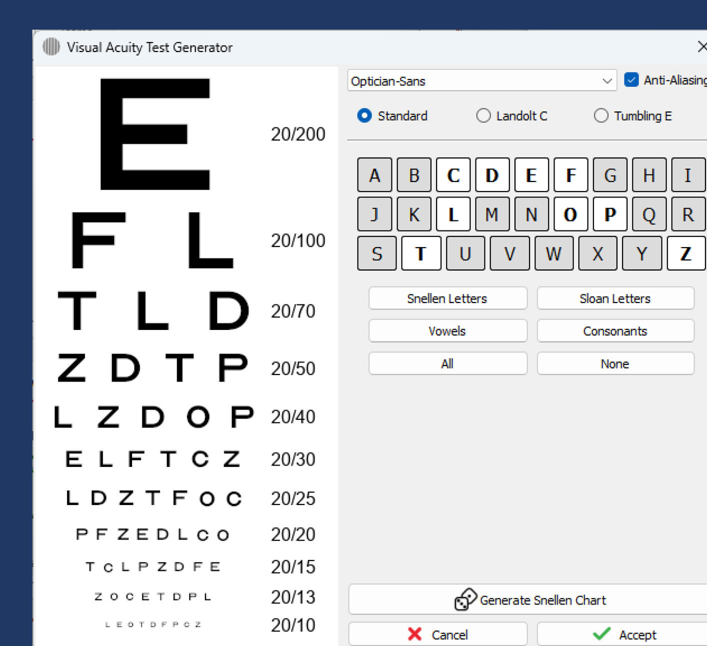
RECOIL 1.3.0 offers a rapid, flexible simulation for visual acuity testing. Key features include:

- **Customizable Acuity Charts**: Generate random charts using Snellen, Sloan, Landolt-C, or Tumbling-E optotypes
- **Optotype Variety**: Choose from 6 different optotype styles
- **Realistic Distortion**: Simulate visual distortions by applying user-defined or interferogram-fitted Zernike polynomials to individual characters
- **Advanced Acuity Assessment**: Determine acuity on a per-character and per-line basis using 7 classifier and aggregate models
- **Comprehensive Results**: View acuity results per model or as an ensembled result; view what humans see through a lens

RECOIL informs lens design in accordance with USAF distortion and vision standards

The Virtual Snellen Acuity Test Process

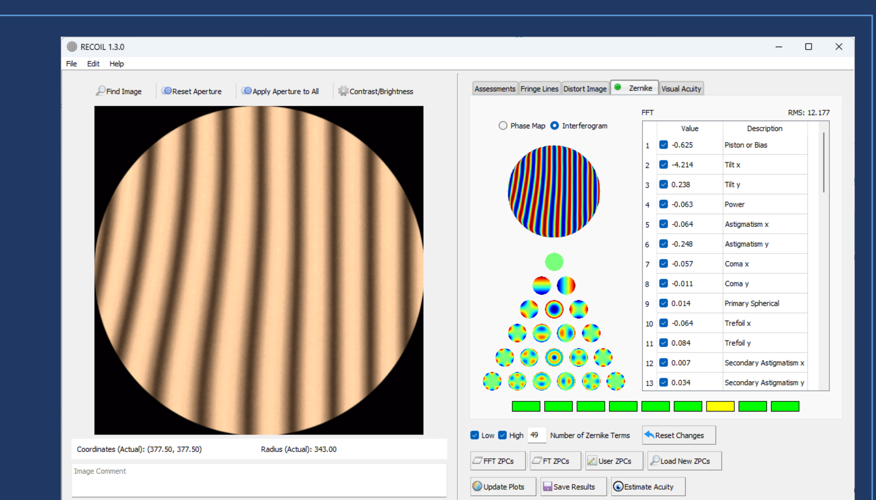




RECOIL's Acuity Chart Generator

RECOIL fits Zernike Polynomials to an interferogram. Users can

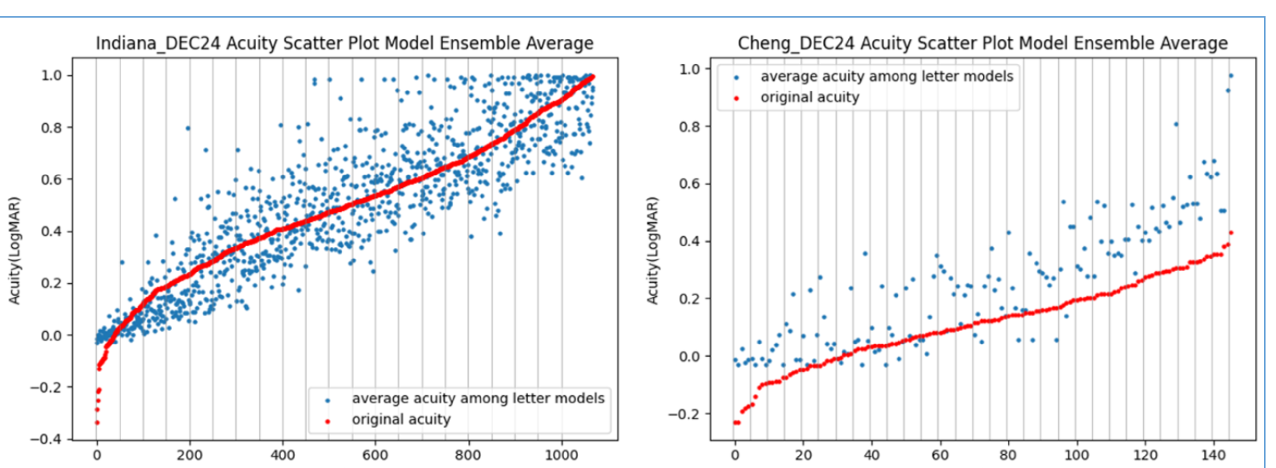
- Adjust the brightness and contrast of the interferogram to improve fitting
- Modify RECOIL's traces or define their own
- Override any fitted terms and visualize the modified wavemap and reconstructed interferogram



RECOIL's Fit of Zernike Polynomials

Validation

- Visual acuity data was obtained from Indiana University and the "Predicting Subjective Judgment of Best Focus" 2004 report in the Journal of Vision by X. Cheng, A. Bradley, and L. Thibos
- Both datasets included Zernike polynomial coefficients (**ZPCs**)
- Visual acuities were predicted using 12 charts: 5 using random Snellen letters, 5 using random Sloan letters, 1 using Landolt-C characters, and 1 using Tumbling-E characters
- The classifier models could only predict models with LogMar acuities between 0 and 1.0



Discussion

- The Cheng dataset had some drawbacks: it was small and only included people with better than 20/20 vision; our models simulated 20/20 vision
- The Indiana dataset didn't have visual acuity values, so we estimated them using a statistical calculation based on the point spread function that correlated with visual acuity
- Even though our models and a human observer typically agreed on acuity chart performance, the acuity values from the Cheng and Indiana datasets often differed greatly from the reported and derived values
- RECOIL's virtual acuity exam provides predictions of visual acuity and lens clarity in a fraction of the time needed to conduct a human study

Acknowledgements & Contact Info

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For more information about this project, please contact Dr. Harvey at aharvey@ara.com or scan the QR code.